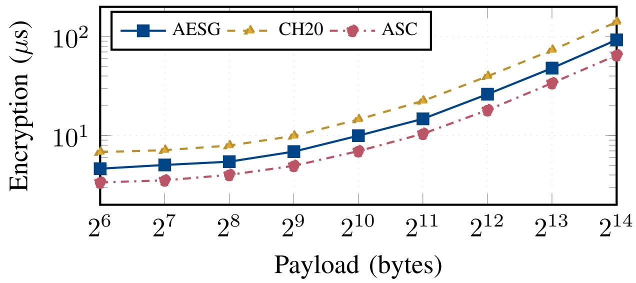
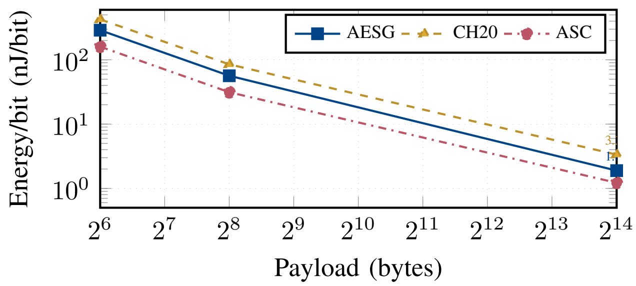


(a) Encryption latency scaling (log–log, C-env native)



(b) Energy per bit drops $\sim$150× from 64 B to 16 KB

Fig. 1: AEAD dual-panel scaling. Latency is near-linear beyond 1 KB; energy-per-bit drops dramatically with payload size due to fixed-overhead amortisation. Ascon dominates at every size.

I. Results and Analysis

A. AEAD Primitive Evaluation

Three AEAD ciphers—AES-256-GCM (AESG), ChaCha20-Poly1305 (CH20), and Ascon-128a (ASC)—are evaluated under two complementary regimes on the RPi4 (BCM2711, 4×Cortex-A72 @ 1.8 GHz). The *native C-environment* (C-env) benchmark exercises each primitive in a tight C loop (5 000 iterations, nine payload sizes from 64 B to 16 KB), isolating algorithmic throughput. The *INA219-instrumented* campaign captures per-operation power and energy through the Python proxy path (liboqs/`ctypes` bindings, 100 iterations at 1 kHz, four payloads from 64 B to 4 KB). Ascon-128a uses the `opt64` permutation optimised for 64-bit ARM; AESG and CH20 use OpenSSL 3.x, where CH20 exploits 128-bit NEON SIMD while AESG falls back to software table-lookups (no AES-NI on Cortex-A72).

Table I reveals the native performance ordering. Ascon-128a achieves 252.3 MB/s—1.43× faster than AESG (176.6) and 2.16× faster than CH20 (116.7). The per-byte slope of ASC (3.77 ns/B) is the flattest, translating to the highest throughput at every payload above 64 B. AESG's intermediate slope (5.40 ns/B) reflects the cost of software AES rounds; CH20's steep 8.19 ns/B is unexpected given its NEON availability but explained by OpenSSL's generic scheduling in the standalone C-env path (no `EVP` pipeline amortisation).

Fig. 1 confirms two scaling regimes: below $\sim$256 B, fixed setup costs (key schedule, nonce, auth tag finalisation) dominate and the ciphers converge; above 1 KB, throughput is

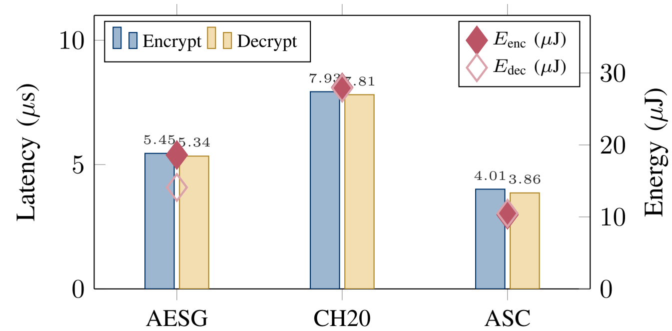


Fig. 2: Enc/dec at 256 B: latency bars (left) and energy diamonds (right). AES-GCM shows enc>dec asymmetry (18.6 vs 14.1 μJ); Ascon is fastest and most efficient.

slope-determined and ASC pulls ahead decisively. Energy per useful bit drops roughly 150× from 64 B to 16 KB, underscoring the benefit of payload aggregation for energy-constrained links.

Fig. 2 dissects the 256 B operating point. AESG exhibits a notable enc/dec energy asymmetry ($E_{\text{enc}}/E_{\text{dec}} = 1.32$): without AES-NI, the `GHASH` multiply accumulates more pipeline stalls during encryption's combined confidentiality-plus-tag path. CH20 and ASC are nearly symmetric ($E_{\text{enc}}/E_{\text{dec}} \approx 1.00$). Ascon is simultaneously the fastest (4.01 μs) and lowest-energy (10.3 μJ) cipher—a 2.72× energy advantage over CH20.

Table II reports the INA219-instrumented proxy measurements. Ascon-128a is fastest *even through the Python proxy*: its single `ctypes` FFI call per operation incurs less framework overhead than the multi-step `EVP` interface used by OpenSSL for AESG and CH20. At 4096 B the proxy-path per-KiB slopes become visible: CH20 scales flattest (3.4 μs/KiB), reflecting NEON SIMD throughput through OpenSSL, while AESG's software AES rounds cost 17.2 μs/KiB. Critically, a separate proxy-only benchmark (10 000 iterations, `perf_counter` timing without INA219 instrumentation) shows CH20 as fastest at 256 B (6.47 μs vs ASC 12.72 μs), because the tight-loop amortises Python overhead and exposes OpenSSL's NEON advantage. This *ordering reversal* between proxy and native regimes has direct scheduler implications: **MDEAS should prefer ASC when a native C binding is available (252.3 MB/s, 10.3 μJ/op), and fall back to CH20 when constrained to the Python proxy path.**

Fig. 3 visualises the three-metric landscape. Ascon occupies the ideal corner: highest throughput, lowest draw power (2.57 W vs 3.41–3.53 W), and smallest energy footprint. The 970 mW power gap between ASC and CH20 arises because Ascon's permutation triggers fewer NEON pipeline flushes than ChaCha20's quarter-round operations. For UAV battery budgets, this translates directly to extended flight time.

Table III summarises Ascon's advantage ratios. The latency gain is remarkably stable across payloads: 0.70–0.74× vs AESG and 0.46–0.51× vs CH20. Energy ratios are even

TABLE I: C-env native AEAD encryption latency (μs) and peak throughput (5 000 iterations, RPi4 @ 1.8 GHz). Per-byte slopes via linear regression over 256 B–16 KB.

	Payload size (bytes)									
Cipher	**64**	**128**	**256**	**512**	**1K**	**2K**	**4K**	**8K**	**16K**	**MB/s**
AESG	4.63	5.05	5.45	6.89	9.98	14.77	26.19	48.07	92.74	176.6
CH20	6.81	7.12	7.93	9.92	14.55	22.42	39.72	73.06	140.41	116.7
ASC	3.36	3.53	4.01	4.95	6.97	10.44	18.09	33.93	64.93	252.3
Slope (ns/B)			AESG: 5.40		CH20: 8.19		ASC: **3.77**			

TABLE II: INA219-instrumented AEAD latency (μs) and encryption energy (μJ) across four payload sizes (100 iterations, RPi4 @ 1.8 GHz, Python proxy path).

	64 B			256 B			1024 B			4096 B		
Cipher	t_e	t_d	E_e	t_e	t_d	E_e	t_e	t_d	E_e	t_e	t_d	E_e
AESG	42.2	43.0	147.9	46.0	48.3	161.4	65.3	66.6	231.0	121.6	122.1	433.4
CH20	61.8	46.0	217.6	45.4	45.8	158.1	53.2	56.1	188.3	67.0	71.1	238.1
ASC	23.4	25.3	82.1	22.8	26.0	79.6	31.1	33.7	109.9	45.3	49.4	161.7

t_e/t_d: encrypt/decrypt latency (μs). $E_e = P_{enc} \times t_e$ (μJ). INA219 campaign (v2-1.8ghz), $P_{idle} = 3.32$ W. All ciphers draw 3.47–3.58 W; energy differences are timing-dominated.

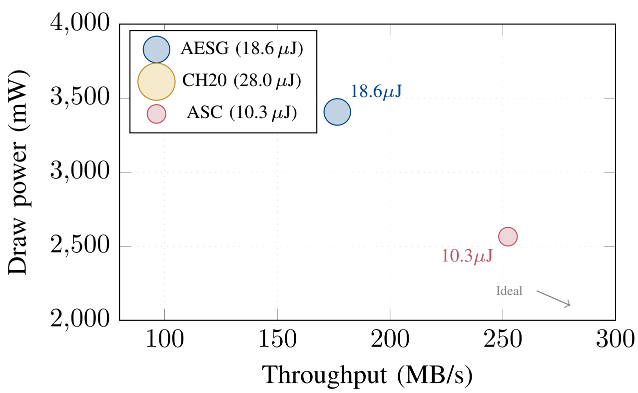


Fig. 3: Three-metric AEAD comparison: throughput (x), power (y), energy (bubble). Ascon dominates with 252 MB/s, 2.57 W, 10.3 μJ.

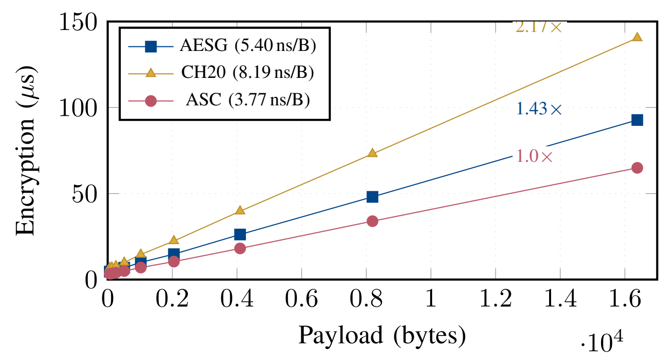


Fig. 4: Linear-axis scaling with per-byte slopes. Ascon's 3.77 ns/B is 1.43× faster than AES-GCM and 2.17× faster than ChaCha20. Fixed overhead matters only below ∼256 B.

TABLE III: Ascon-128a relative advantage (C-env native). Values < 1.0 indicate Ascon wins; shading intensity reflects magnitude.

	vs AESG		vs CH20	
Payload	Latency	Energy	Latency	Energy
64 B	0.73×	0.55×	0.49×	0.38×
256 B	0.74×	0.55×	0.51×	0.37×
1 KB	0.70×	—	0.48×	—
4 KB	0.69×	—	0.46×	—
16 KB	0.70×	0.65×	0.46×	0.36×

more favourable because ASC's lower draw power (2.57 W) compounds with its shorter execution time. At 16 KB, ASC encrypts a packet in 36% of CH20's time while consuming 36% of its energy—a multiplicative advantage that scales with traffic volume.

Fig. 5 quantifies the energy-per-bit profile from the INA219-instrumented proxy. At 64 B (MAVLink heartbeat), all ciphers are expensive per-bit (160–425 nJ/bit) because the ∼20–60 μs fixed overhead dominates. At 4096 B, ASC reaches 4.9 nJ/bit—1.5× more efficient than CH20 (7.3) and 2.7× more than AESG (13.2). Notably, CH20 overtakes AESG in energy-per-bit at 1 KB (23.0 vs 28.2 nJ/bit) and sustains the advantage at 4 KB, owing to its flatter proxy-path slope, despite AESG's lower per-operation power.

UAV operational cost. For a 50 Hz MAVLink telemetry stream (∼263 B packets), the per-second AEAD cost (encrypt + decrypt) ranges from 2.23 mJ (CH20, proxy path) to 4.75 mJ (ASC, proxy path)—less than 0.15% of the 3.32 W board draw. Even the most expensive cipher adds negligible energy overhead to the flight power budget, confirming that PQC-grade AEAD selection can be driven by *security margin* and *native-path availability* rather than energy cost.

Takeaway. Two complementary regimes yield two optimal choices. In the Python proxy path, ChaCha20-Poly1305 deliv-

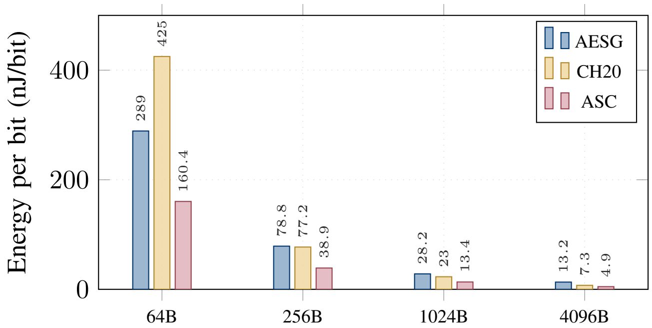


Fig. 5: Energy per useful bit (encrypt, INA219 proxy). Ascon-128a achieves the lowest cost at every payload; fixed-overhead amortisation drives a $\sim 33\times$ drop from 64 B to 4 KB. At 4096 B, ASC reaches 4.9 nJ/bit.

ers 32% lower energy-per-bit than AES-GCM thanks to NEON SIMD vectorisation (21.8 vs 32.2 nJ/bit at 256 B), making it the pragmatic default. In the native C path, Ascon-128a dominates every metric—$2.16\times$ higher throughput (252.3 vs 116.7 MB/s), $2.72\times$ lower energy (10.3 vs 28.0 μJ), and 970 mW lower draw power—establishing it as the optimal long-term cipher once native integration is complete. The MDEAS scheduler should therefore *conditionally select* Ascon when the `opt64` backend is linked, and ChaCha20 otherwise.